

Long Read Assembly

Michael Schatz

Sept 16, 2026

Lecture 5: Applied Comparative Genomics



Assignment I

github.com/schatzlab/appliedgenomics2026/blob/main/assignments/assignment1/README.md

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tomato.chrom.sizes

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README.md

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mschatz add draft of hw1 14d4cd0 · 9 minutes ago History

Preview Code Blame 195 lines (120 loc) · 11.4 KB

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Assignment 1: Chromosome Structures

Assignment Date: Wednesday, September 9, 2026
Due Date: Wednesday, Sept. 16, 2026 @ 11:59pm

Assignment Overview

In this assignment you will profile the overall structure of the genomes of several important species and then study human chromosome 22 in more detail. As a reminder, any questions about the assignment should be posted to [Piazza](#).

Question 1: Why Genomics? [10 pts]

- Question 1.1 Use ChatGPT (or your favorite LLM) to write an essay on why you are interested in genomics. Include specific details from your own interests. Make sure to ask for references. Make sure to include both your prompt(s) and the output from the LLM
- Question 1.2 Comment on the output from the LLM - does it make logical sense, does it include any phrases you would not have written, do the cited papers exist and support the claims from the LLM?
- Question 1.3 Post your final prompt, the LLM's response, and which LLM you used to piazza. Include the link to the post in your submission

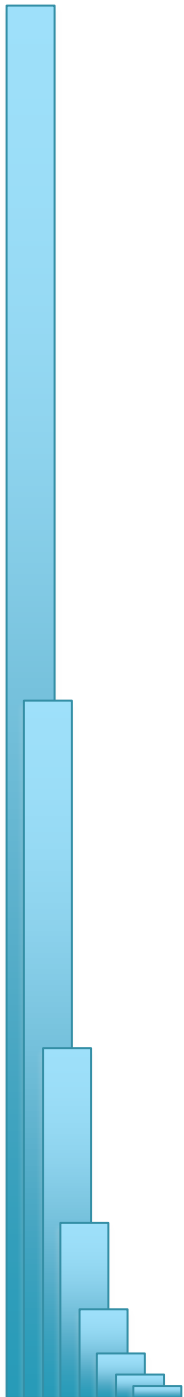
Question 2: Chromosome structures [10 pts]

Download the chromosome size files for the following genomes (Note these have been preprocessed to only include main chromosomes):

- [E. coli \(Escherichia coli K12\)](#) - One of the most commonly studied bacteria [\[info\]](#)
- [Yeast \(Saccharomyces cerevisiae, sacCer3\)](#) - an important eukaryotic model species, also good for bread and beer [\[info\]](#)
- [Worm \(Caenorhabditis elegans, ce10\)](#) - One of the most important animal model species [\[info\]](#)
- [Fruit Fly \(Drosophila melanogaster, dm6\)](#) - One of the most important model species for genetics [\[info\]](#)
- [Arabidopsis thaliana \(TAIR10\)](#) - An important plant model species [\[info\]](#)
- [Tomato \(Solanum lycopersicum v4.00\)](#) - One of the most important food crops [\[info\]](#)
- [Human \(hg38\) - us :](#) [\[info\]](#)
- [Wheat \(Triticum aestivum, IWGSC\)](#) - The food crop which takes up the largest land area [\[info\]](#)

Using these files, make a table with the following information per species (Recall the N50 length is the length such that 50% of the

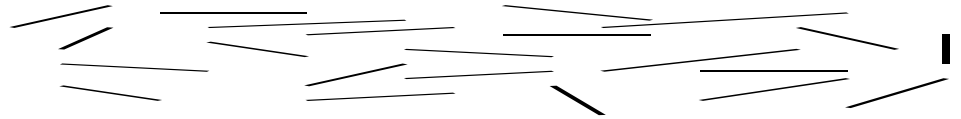
<https://github.com/schatzlab/appliedgenomics2026/tree/main/assignments/assignment1>
Due end of day on Wednesday Sept 16 (right before midnight)



0. Recap

Assembling a Genome

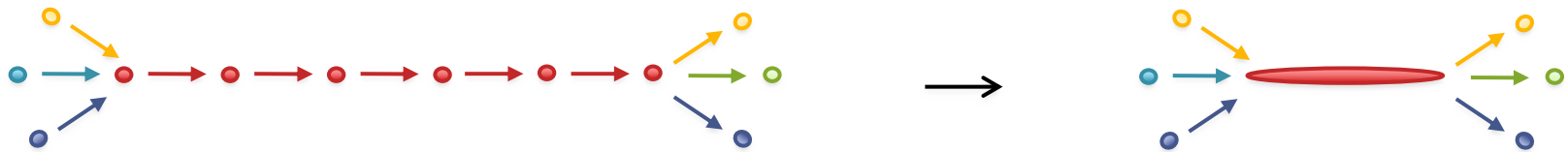
1. Shear & Sequence DNA



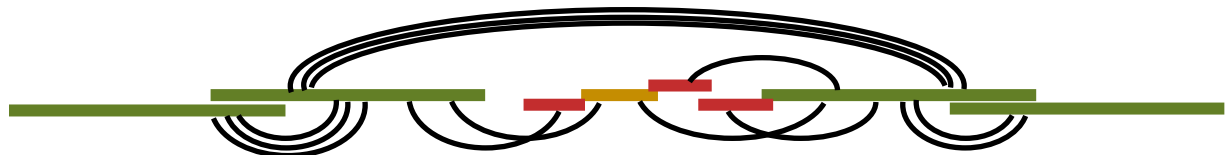
2. Construct assembly graph from reads (de Bruijn / overlap graph)

...AGCCTAGGGATGCGCGACACGT
GGATGCGCGACACGT CGCATATCCGGTTTGGT CAACCTCGGACGGAC
CAACCTCGGACGGAC CTCAGCGAA...

3. Simplify assembly graph

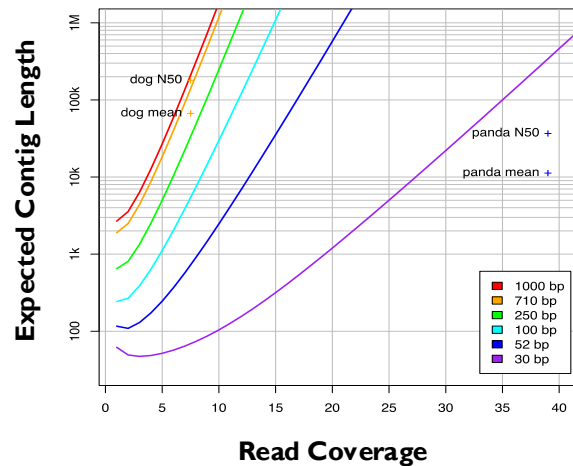


4. Detangle graph with long reads, mates, and other links



Ingredients for a good assembly

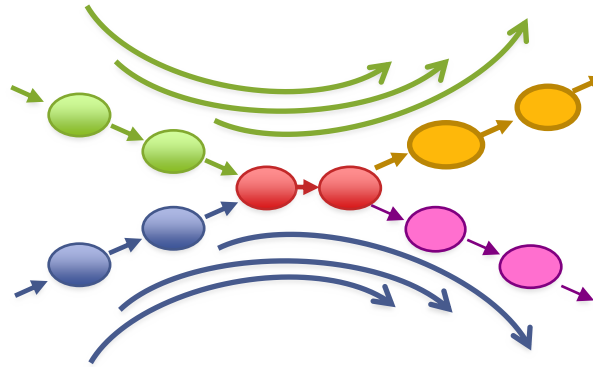
Coverage



High coverage is required

- Oversample the genome to ensure every base is sequenced with long overlaps between reads
- Biased coverage will also fragment assembly

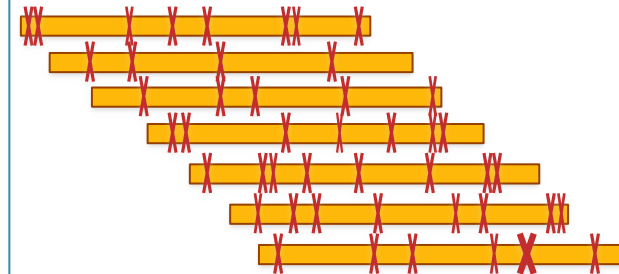
Read Length



Reads & mates must be longer than the repeats

- Short reads will have **false overlaps** forming hairball assembly graphs
- With long enough reads, assemble entire chromosomes into contigs

Quality



Errors obscure overlaps

- Reads are assembled by finding kmers shared in pair of reads
- High error rate requires very short seeds, increasing complexity and forming assembly hairballs

Current challenges in *de novo* plant genome sequencing and assembly

Schatz MC, Witkowski, McCombie, VWR (2012) *Genome Biology*. 12:243

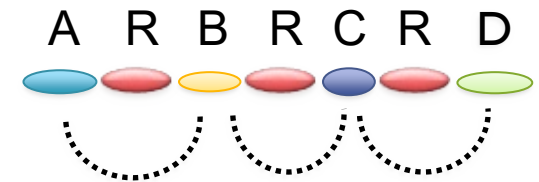
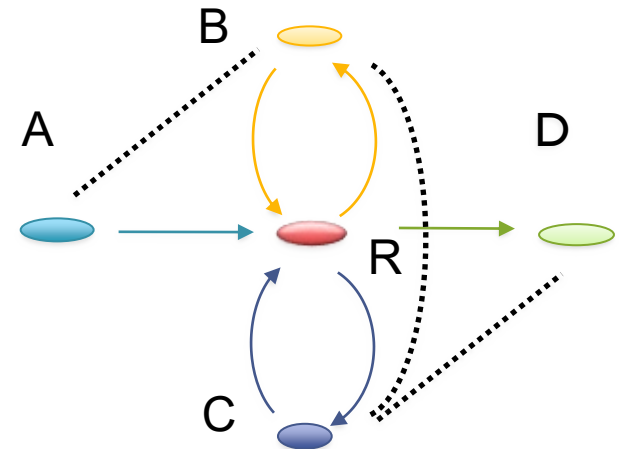
Repetitive regions

Repeat Type	Definition / Example	Prevalence
Low-complexity DNA / Microsatellites	$(b_1b_2\dots b_k)^N$ where $1 \leq k \leq 6$ CACACACACACACACACA	2%
SINEs (Short Interspersed Nuclear Elements)	<i>Alu</i> sequence (~280 bp) Mariner elements (~80 bp)	13%
LINEs (Long Interspersed Nuclear Elements)	~500 – 5,000 bp	21%
LTR (long terminal repeat) retrotransposons	Ty1-copia, Ty3-gypsy, Pao-BEL (~100 – 5,000 bp)	8%
Other DNA transposons		3%
Gene families & segmental duplications		4%

- Over 50% of mammalian genomes are repetitive
 - Large plant genomes tend to be even worse
 - Wheat: 16 Gbp; Pine: 24 Gbp

Scaffolding

- Initial contigs (*aka* unipaths, unitigs) terminate at
 - Coverage gaps: especially extreme GC
 - Conflicts: errors, repeat boundaries
- Use mate-pairs to resolve correct order through assembly graph
 - Place sequence to satisfy the mate constraints
 - Mates through repeat nodes are tangled
- Final scaffold may have internal gaps called sequencing gaps
 - We know the order, orientation, and spacing,



Why do contigs & scaffolds end?

(1) end of chromosome; (2) gap in coverage; (3) repeats; (4) heterozygosity

1. **Coverage**: low coverage is mathematically hopeless
2. **Repeat composition**: high repeat content is challenging
3. **Read length**: longer reads help resolve repeats
4. **Error rate**: errors reduce coverage, obscure true overlaps

- Assembly is a hierarchical, starting from individual reads, build high confidence contigs/unitigs, incorporate the mates to build scaffolds
 - Extensive error correction is the key to getting the best assembly possible from a given data set
- Recommend spades for short read assembly
 - Integrates error correction and scaffolding



The Sequence of the Human Genome

Venter et al.

Science 291, pp 1304-1351 (2001)



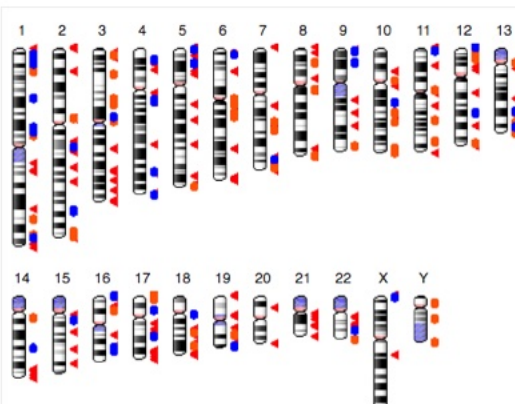
Initial sequencing and analysis of the human genome

International Human Genome Sequencing Consortium

Nature 409, pp 860-921 (2001)

Human Genome Overview

Information about the continuing improvement of the human genome



- ◀ Region containing alternate loci
- Region containing fix patches
- Region containing novel patches

Ideogram of the latest human assembly, GRCh38.p11

The GRC is working hard to provide the best possible representation of the human genome by both generating multiple representations (alternates) and by providing a single path. Additionally, we are releasing alternate assemblies that allows users who are interested in a specific locus to view the data without affecting users who need chromosome coordinate space.

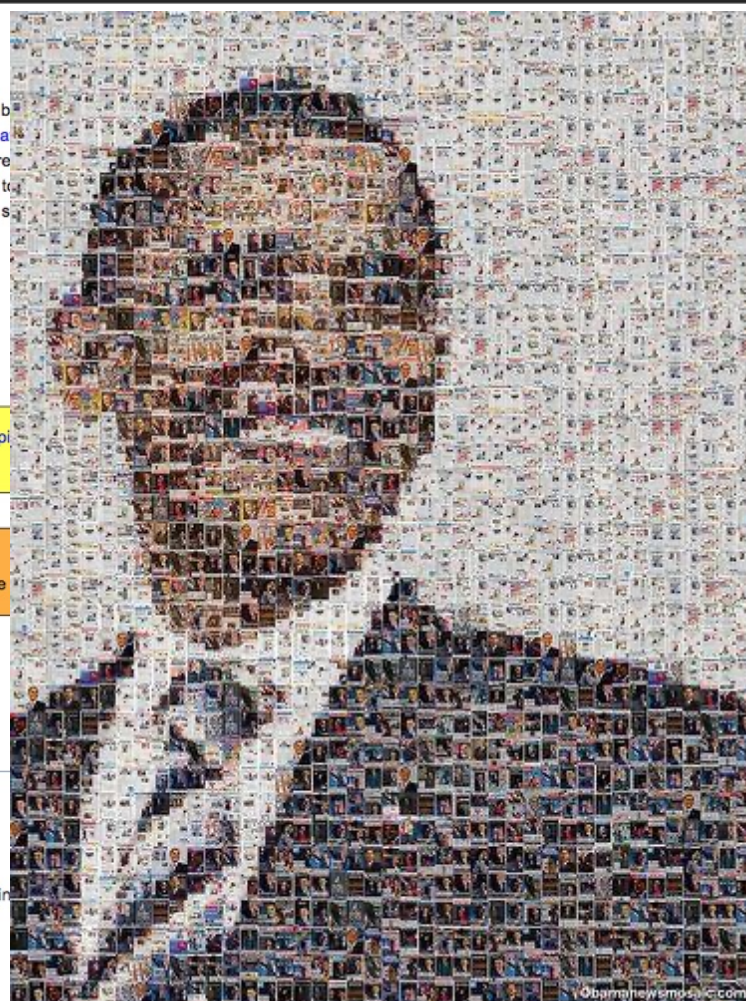
Download data:

- [GRCh38.p11 \(latest minor release\) FTP](#)
- [GRCh38 \(latest major release\) FTP](#)
- [Genomic regions under review FTP](#)
- [Current Tiling Path Files \(TPFs\)](#)

Transitioning to GRCh38? Try the [NCBI Remapping tool](#) to view the assembly alignments used by the GRC.

Next assembly update

The next assembly update (GRCh38.p12) will be available in the near future.


[GRCh38.p11](#)
[GRCh37.p13](#)
[GRCh37](#)

GRCh38.p11

Release date: June 14, 2017

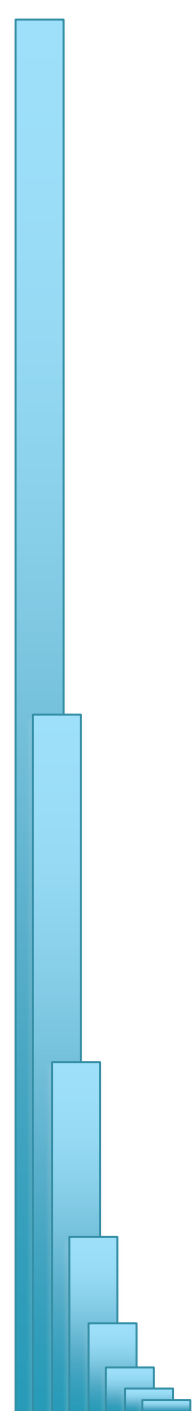
Release type: minor

Release notes: GRCh38.p11 is the eleventh patch release for the GRCh38 reference assembly. No chromosome coordinate changes. The number of patch scaffolds is now: 64 FIX and 59 NOVEL.

Assembly accessions: GenBank: [GCA_000001405.26](#) , RefSeq: [GCF_000001405.37](#)

Pseudoautosomal regions

Name	Chr	Start	Stop
PAR#1	X	10,001	2,781,479
PAR#2	X	155,701,383	156,030,895
PAR#1	Y	10,001	2,781,479
PAR#2	Y	56,887,903	57,217,415

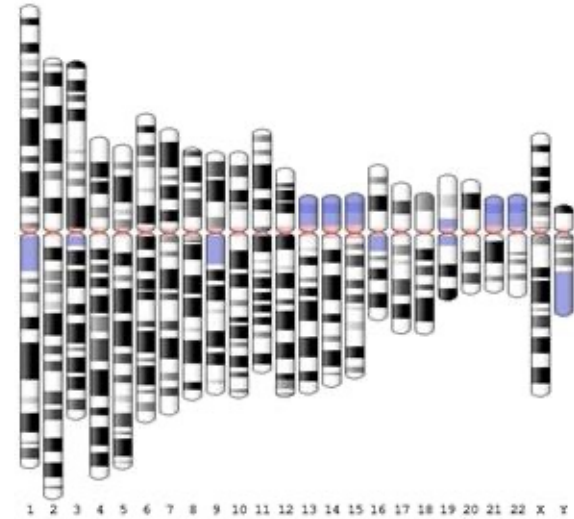


Part I: Long Read Assembly

Finishing the human genome

238Mbp is missing or incorrect

- Centromeres and telomeres
- Segmentally duplicated genes
- Tandem gene arrays (e.g. rDNAs)
- And an unknown number of errors...



Why does it matter?

Variation in these regions is unexplored

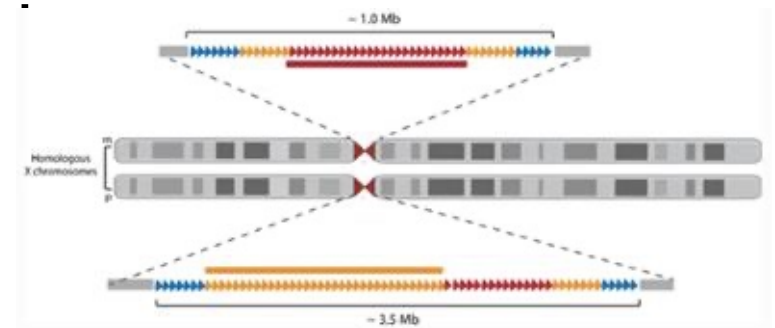
Functional studies need sequence

Reference gaps lead to artifacts

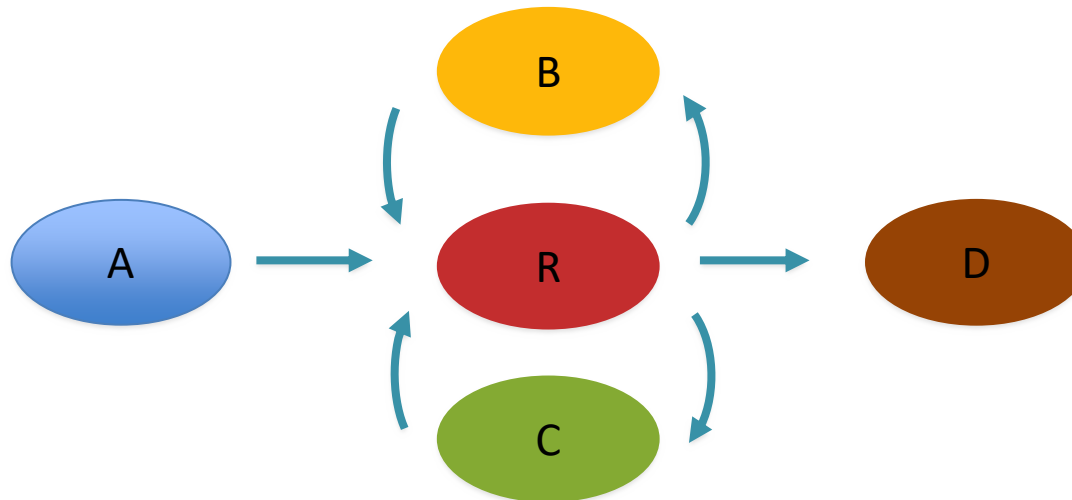
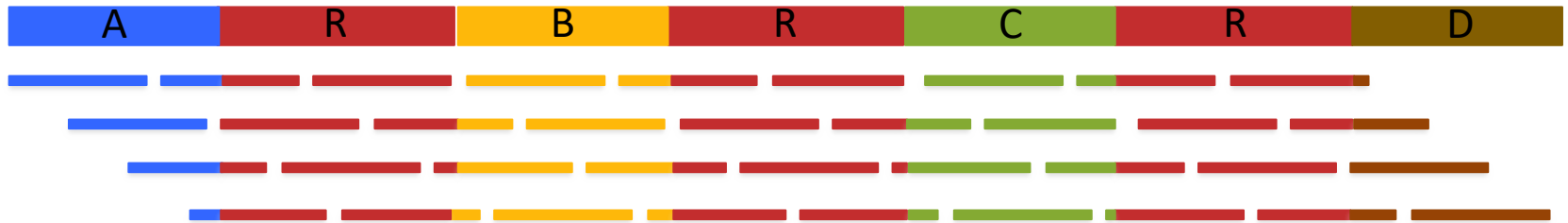
We don't know what we don't

Why has it taken so long?

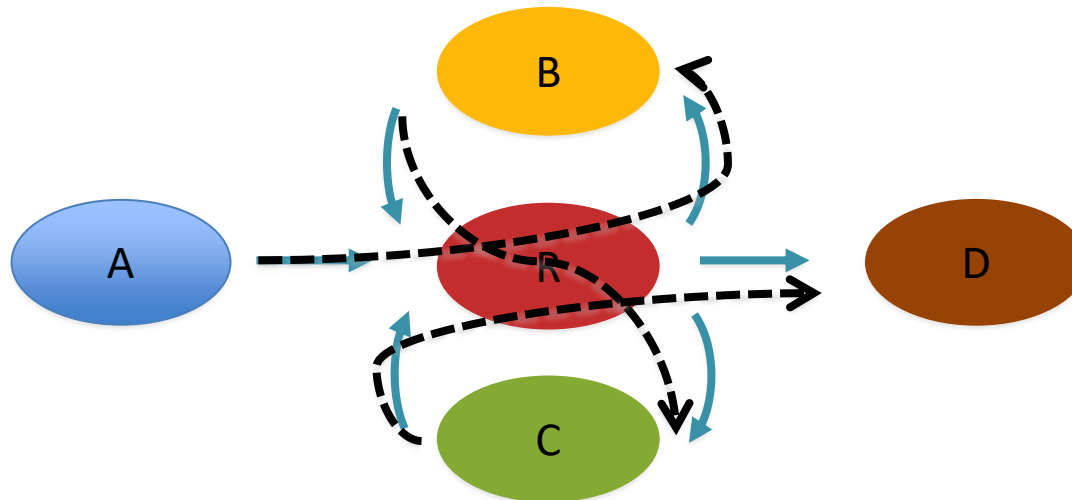
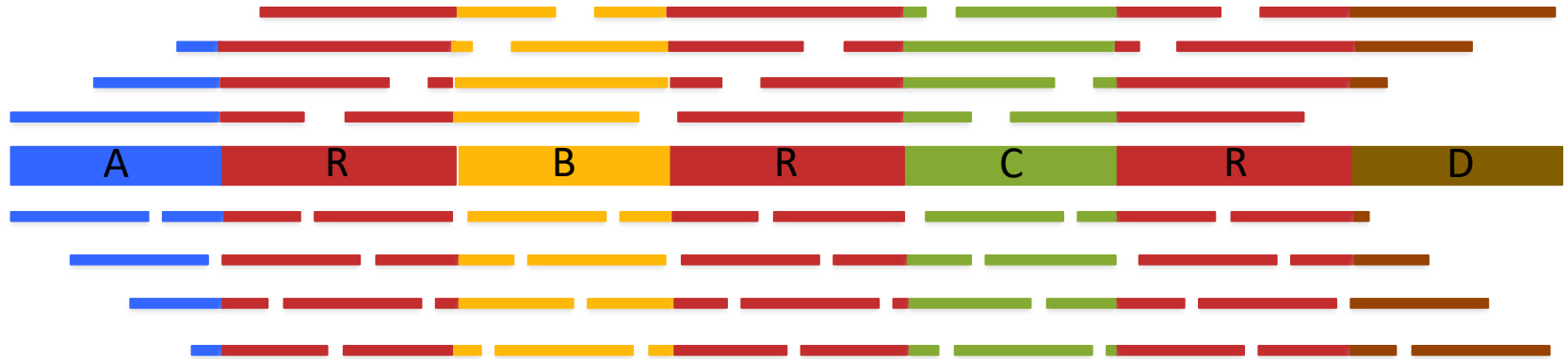
Repeats, repeats, repeats...



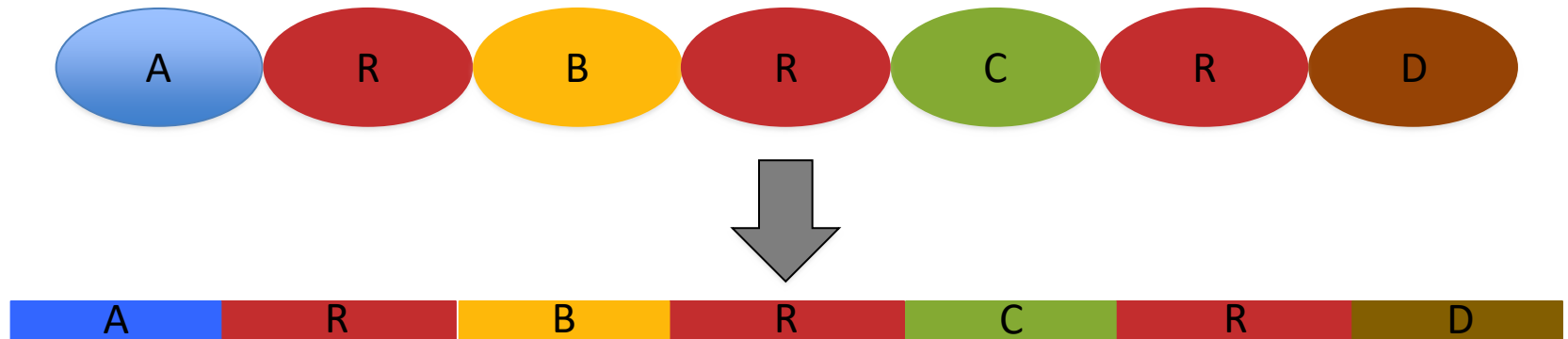
Assembly Complexity



Assembly Complexity



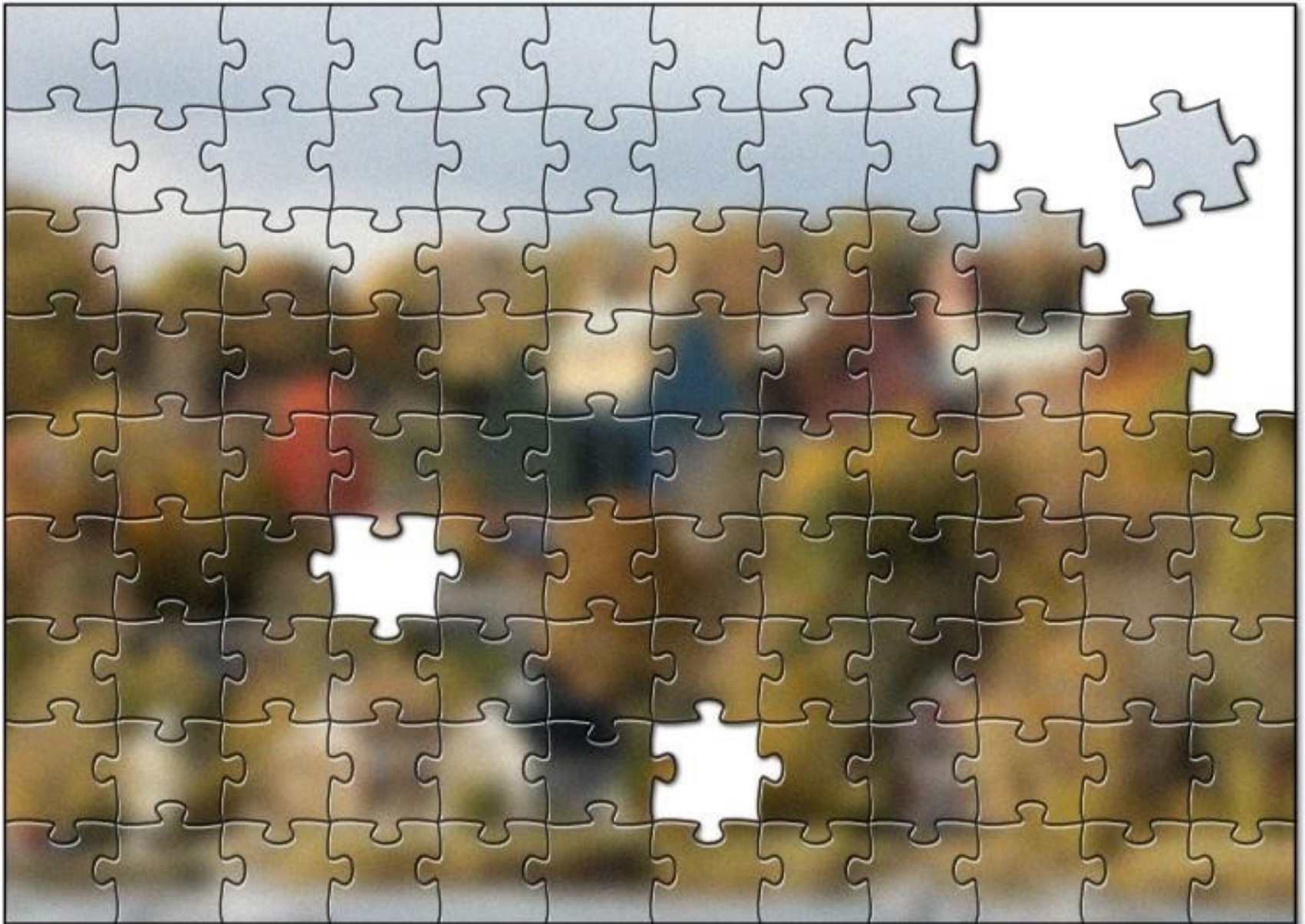
Assembly Complexity



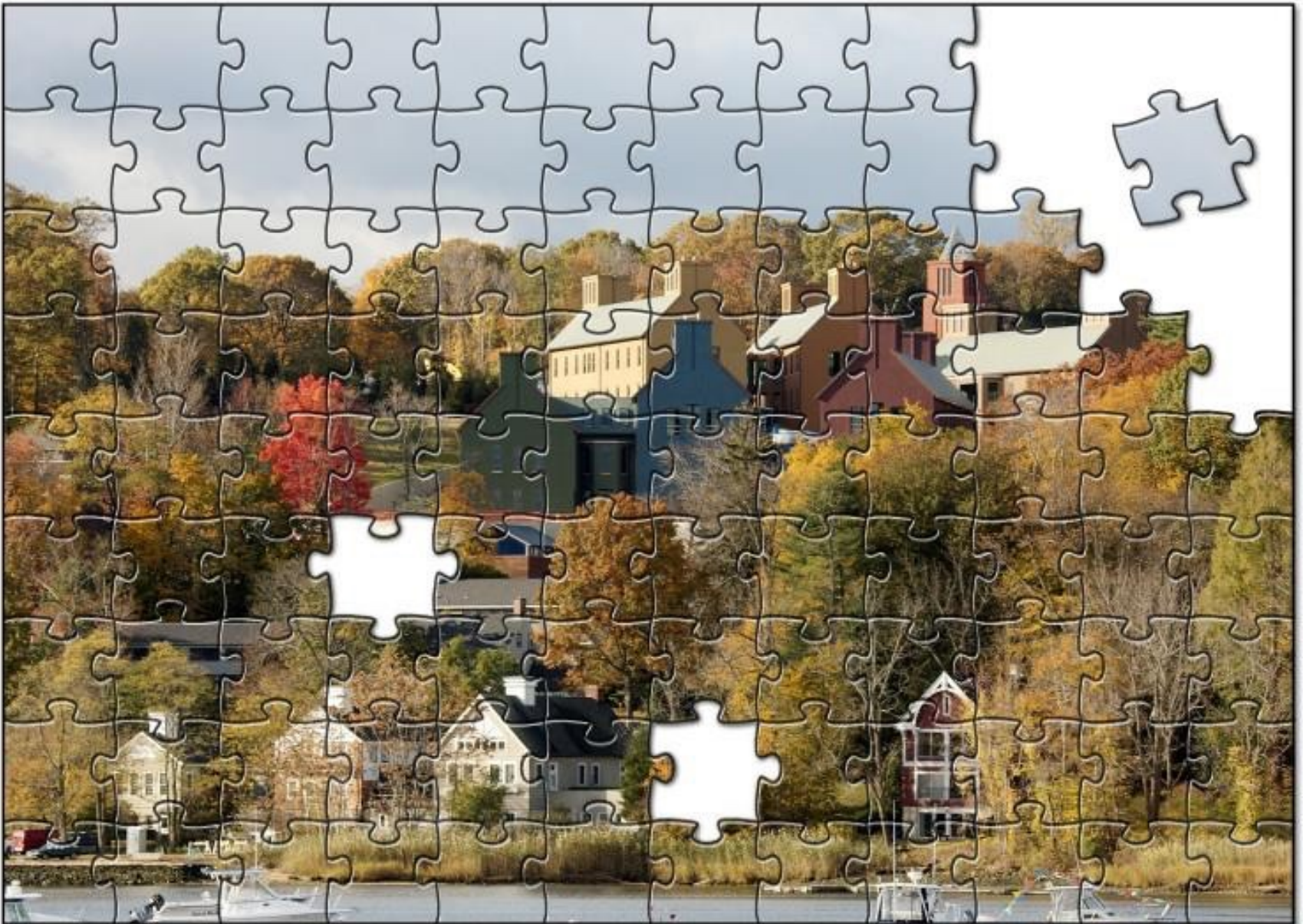
The advantages of SMRT sequencing

Roberts, RJ, Carneiro, MO, Schatz, MC (2013) *Genome Biology*. 14:405

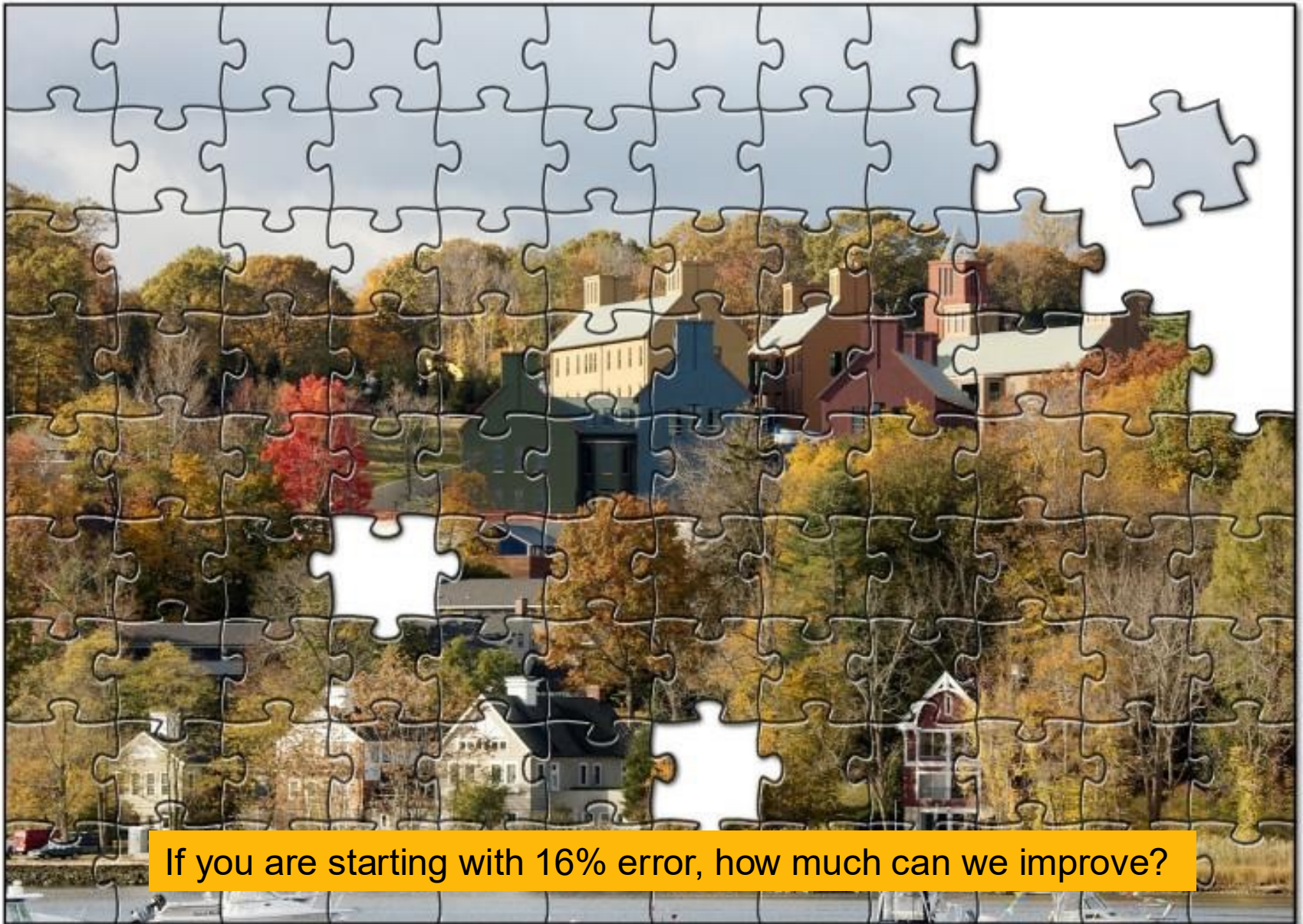
Single Molecule Sequences



“Corrective Lens” for Sequencing

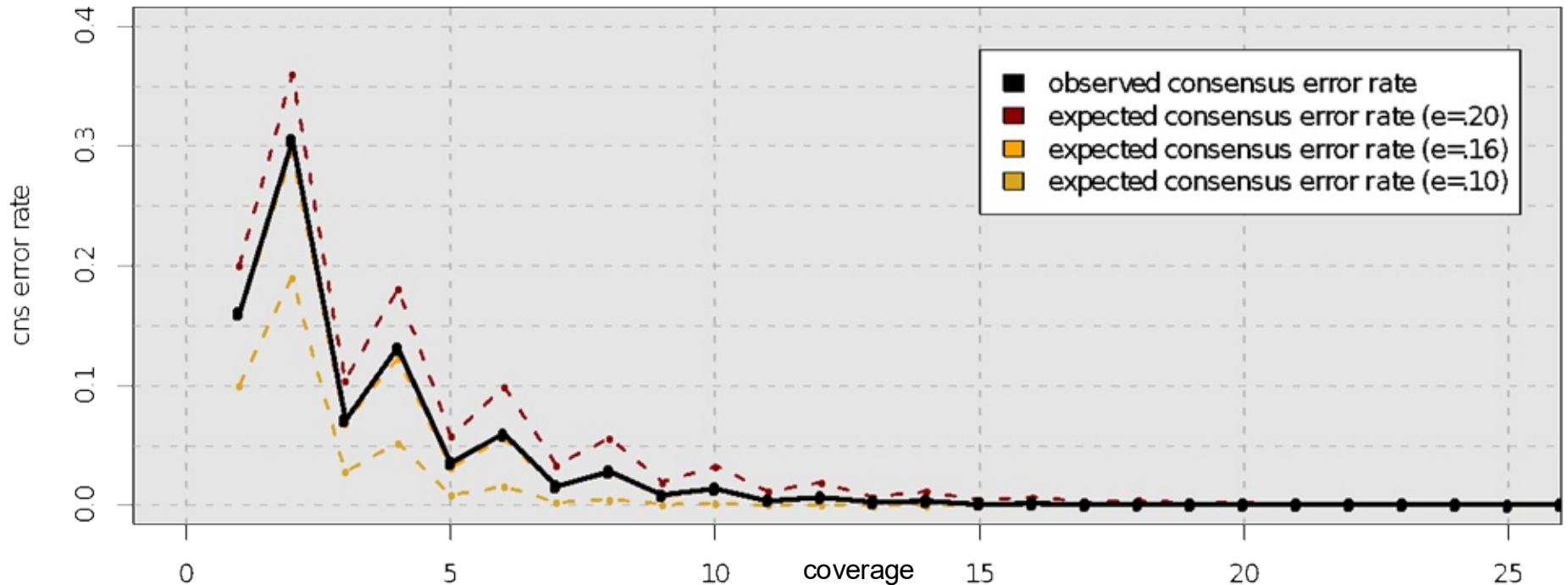


“Corrective Lens” for Sequencing



If you are starting with 16% error, how much can we improve?

Consensus Accuracy and Coverage



Coverage can overcome random errors

- Dashed: error model from binomial sampling; solid: observed accuracy
- For same reason, CCS is extremely accurate when using 5+ subreads

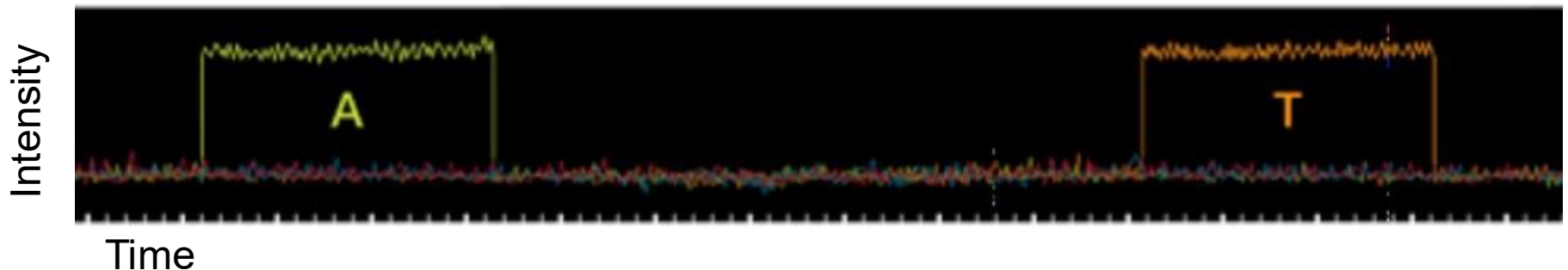
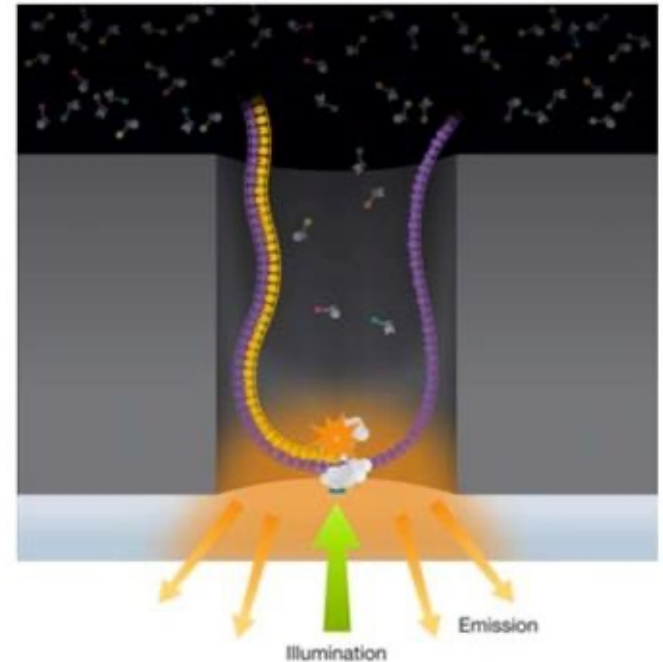
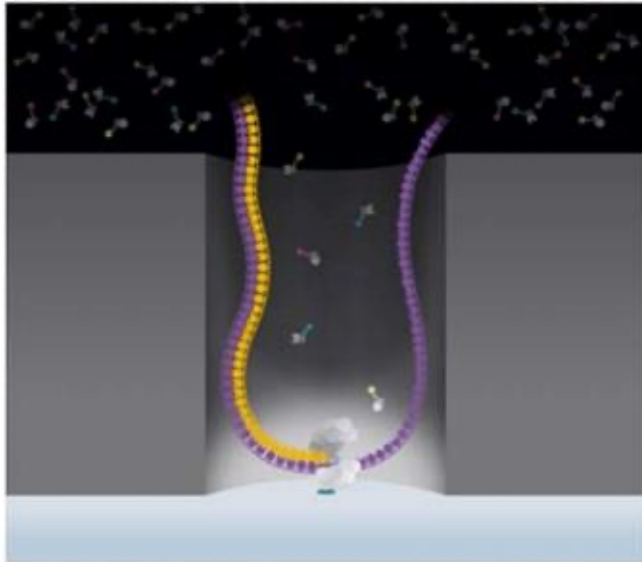
$$CNS\ Error = \sum_{i=\lceil c/2 \rceil}^c \binom{c}{i} (e)^i (1-e)^{n-i}$$



PacBio Single
Molecule Real Time
Sequencing
(SMRT-sequencing)

PacBio: SMRT Sequencing

Imaging of fluorescent phospholinked labeled nucleotides as they are incorporated by a polymerase anchored to a Zero-Mode Waveguide (ZMW).



“HiFi” Circular Consensus Reads

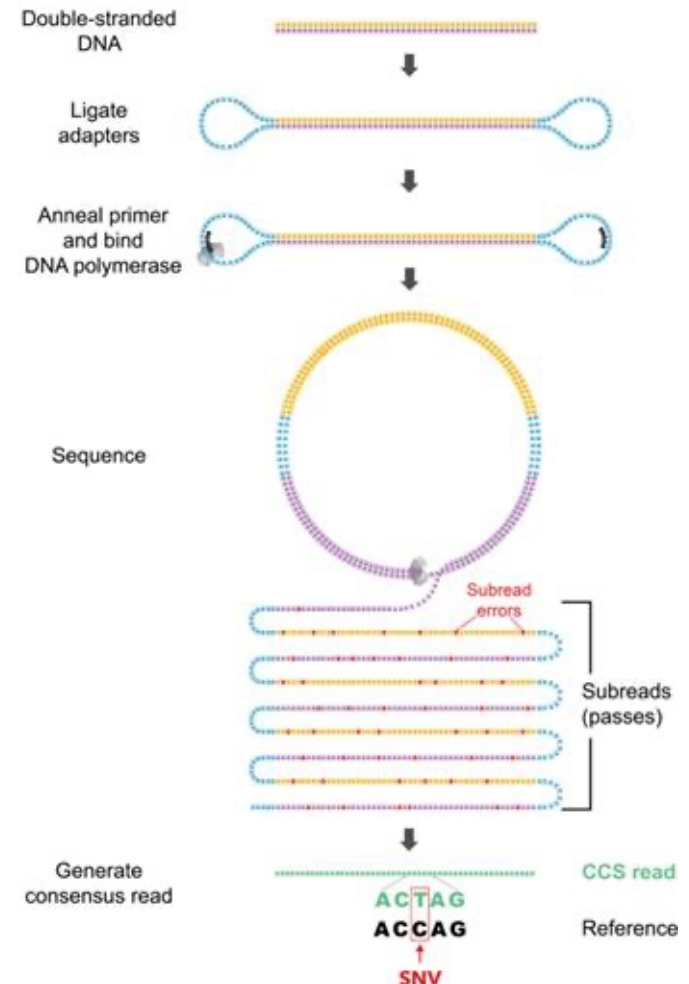
High-quality reads produced by sequencing the same molecule multiple times

Higher accuracy for low-coverage sequences like somatic variants or lowly expressed transcripts in RNA-seq, more interpretable alignments, better & faster assembly

Limits read length, used to be very expensive but more manageable now

Accurate circular consensus long-read sequencing improves variant detection and assembly of a human genome

Wenger et al(2019) Nature Biotechnology doi:10.1038/s41587-019-0217-9

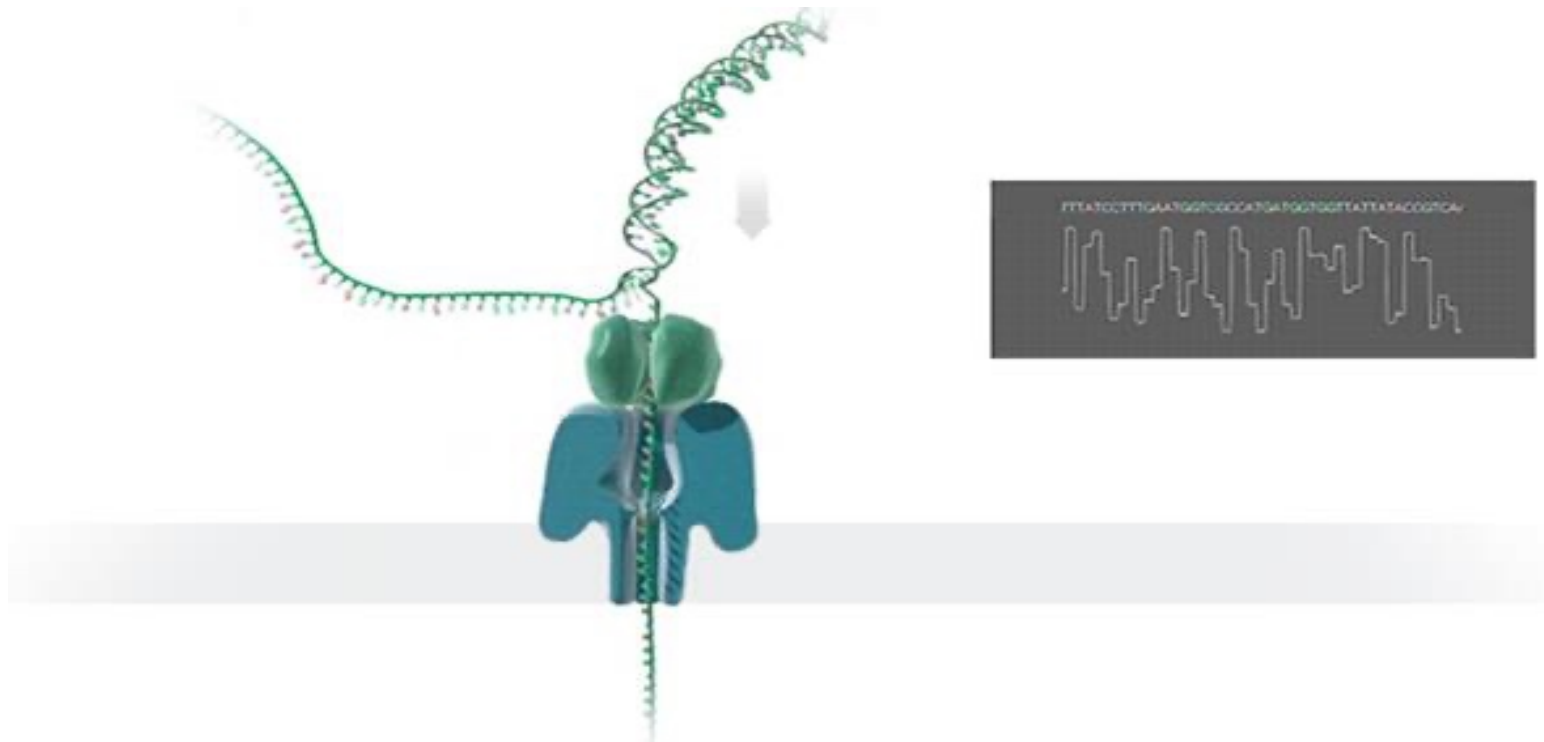


Oxford Nanopore Technologies (ONT)



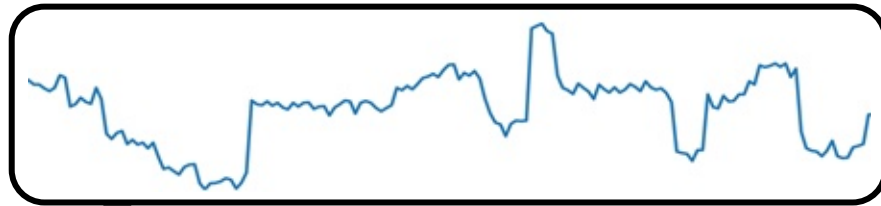
Nanopore Sequencing

Sequences DNA/RNA by measuring changes in ionic current as nucleotide strand passes through a pore

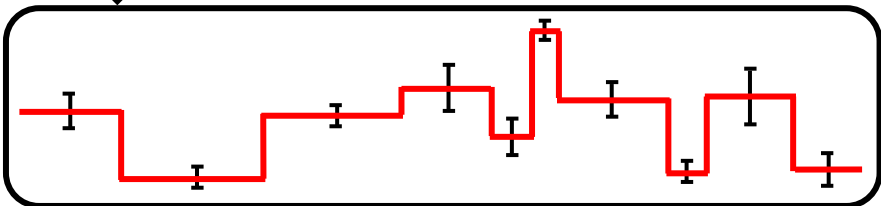


Nanopore Basecalling

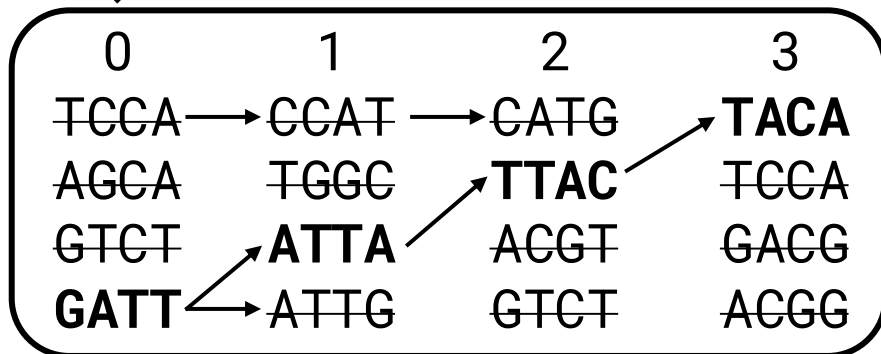
Raw Signal



Events

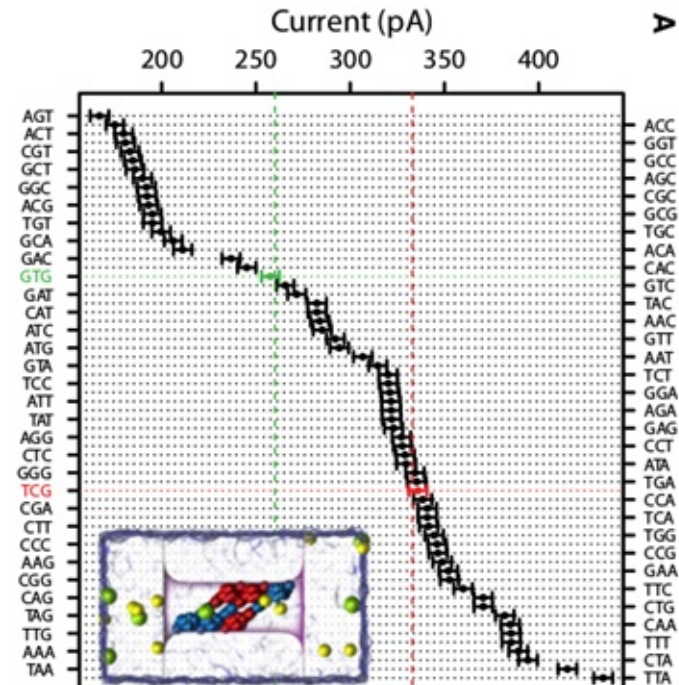


Possible k-mers

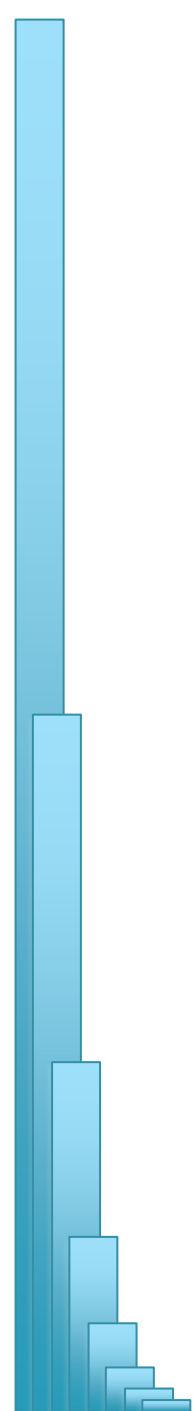


GATTACA

Final sequence determined by most probable k-mers



"DNA Base-Calling from a Nanopore Using a Viterbi Algorithm"
Timp et al. (2012) *Biophysical Journal*



Part 2: The T2T human genome

Let's finish a human genome



T2T Working Group

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[Technology](#)

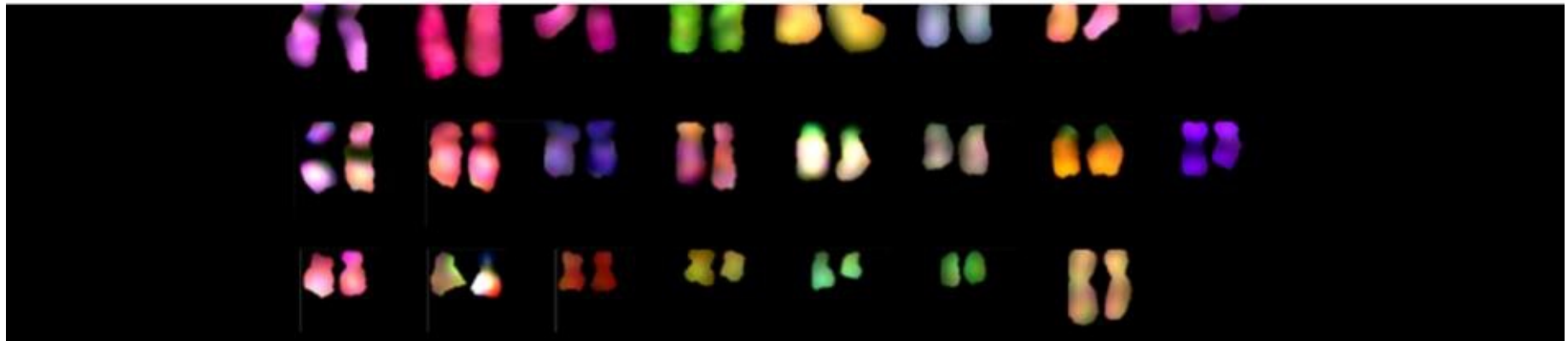
[Data](#)

[CHM13 Cell Line](#)

[Remaining Challenges](#) ▼

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The Telomere-to-Telomere (T2T) consortium is an open, community-based effort to generate the first complete assembly of a human genome.

CHM13 homozygous 46,XX cell line from Urvashi Surti, Pitt; SKY karyotype from Jennifer Gerton, Stowers

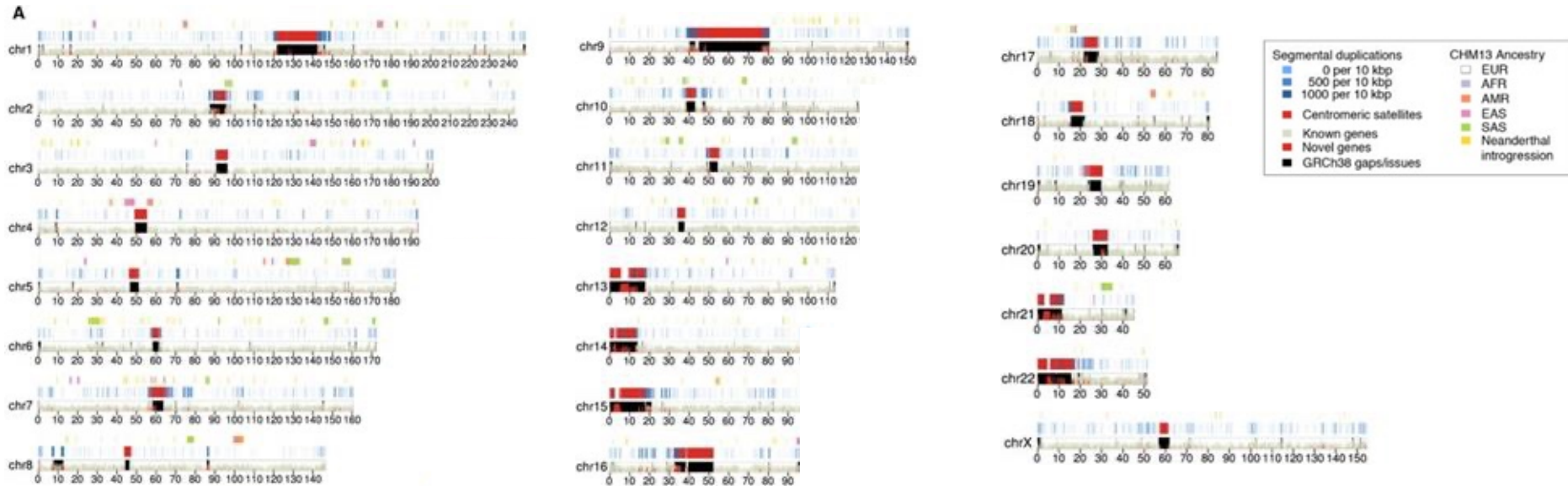
CHM13 assembly graph



HiCanu: accurate assembly of segmental duplications, satellites, and allelic variants from high-fidelity long reads.

Nurk et al. *Genome Research* (2020)

The complete sequence of a human genome



CHM13v1.1 genome size is **3.057 Gbp with zero Ns**

Every chromosome is telomere-to-telomere, quality estimated >Q70
~190 Mbp (~8%) of new sequence vs. GRCh38, fixes thousands of errors

(Nurk et al. Science, 2022)

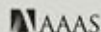
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and relapse pp. 33 & 86

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in a nova explosion p. 77

Science

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SPECIAL ISSUE
science.org



FILLING THE GAPS

Closing in on a complete
human genome p. 42

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LOG IN BECOME A MEMBER

HOME COLLECTIONS COMPLETING THE HUMAN GENOME

COMPLETING THE HUMAN GENOME

A fully sequenced human genome was announced more than 20 years ago. However, owing to technological limitations, some genomic regions remained unresolved. Here, *Science* and other journals present research by the Telomere-to-Telomere (T2T) Consortium, reporting on the endeavor to complete a comprehensive human reference genome.

FILTERS 6 RESULTS FOUND

SPECIAL ISSUE RESEARCH ARTICLE

Segmental duplications and their variation in a complete human genome

BY MITCHELL R. VOLLGER, XAVI GUSTART, PHILIP C. DUBACH, LUDOVICA MERCURI, WILLIAM T. HARVEY, ARIEL GERSHMAN, MARK DEKHANE, ARVIS SILOVAKI, KATHERINE M. MUNDSON, ALEXANDRA P. LEWIS, [...] EVAN E. EICHLE

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Complete genomic and epigenetic maps of human centromeres

BY NICOLAS ALTEMOSE, GLENNIS A. LOGSDON, ANDREY V. BZKAJDE, PRADYK SIKHARAN, SASHA A. LANGLEY, DINA V. CALDAS, SAVANNAH J. HOYT, LEV URALSKY, FEDOR D. RYABOV, COLIN J. SHEW, [...] KAREN H. MIGA

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From telomere to telomere: The transcriptional and epigenetic state of human repeat elements

BY SAVANNAH J. HOYT, JESSICA M. STORER, GABRIELLE A. HARTLEY, PATRICK G. S. GRADY, ARIEL GERSHMAN, LEONARDO G. DE LIMA, CHARLES LAMOUSE, REZA HALABIAN, LIKE WUJENSKI, MATIAS RODRIGUEZ, [...] RACHEL A.

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SPECIAL ISSUE RESEARCH ARTICLE

A complete reference genome improves analysis of human genetic variation

BY SERGEY AGANZOV, STEPHANIE M. YAN, DANIELA C. SOTO, MELANIE KIRSCH, SAMANTHA ZARATE, PAVEL RYDEYEV, DYLAN J. TAYLOR, KISHOR SHAFIN, ALAINA SHUMATE, CHUNLIN XIAO, [...] MICHAEL C. SCHATZ

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SPECIAL ISSUE RESEARCH ARTICLE

Epigenetic patterns in a complete human genome

BY ARIEL GERSHMAN, MICHAEL E. G. SAURIA, XAVI GUSTART, MITCHELL R. VOLLGER, PAUL W. HOOK, SAVANNAH J. HOYT, METEN JIAN, ALAINA SHUMATE, ROHAM RAZAGHI, SERGEY KOREN, [...] WINSTON TAM

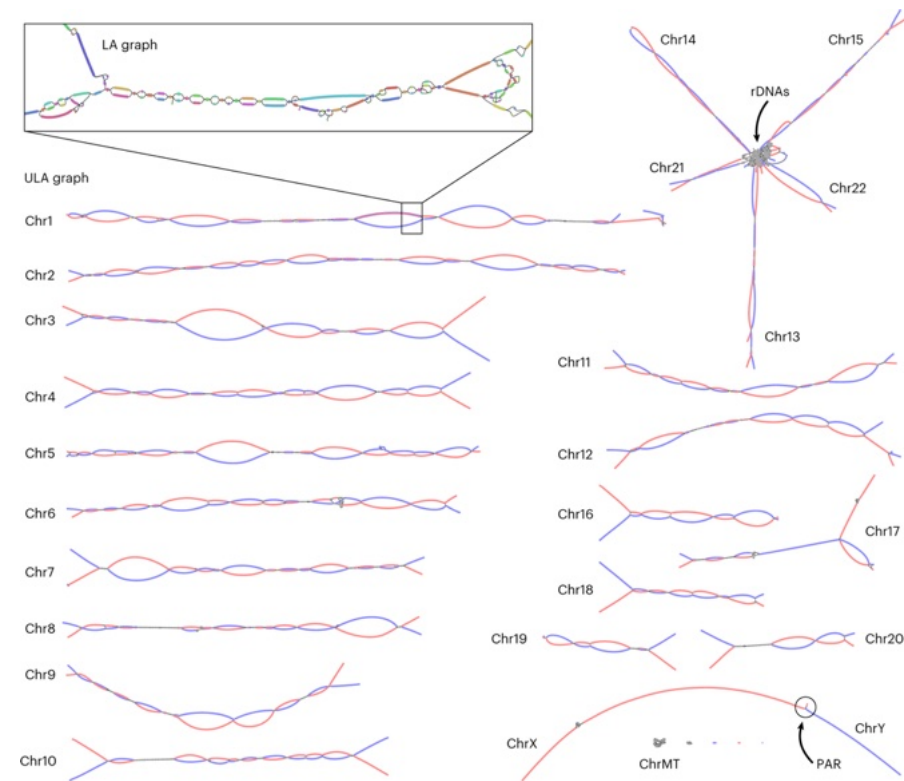
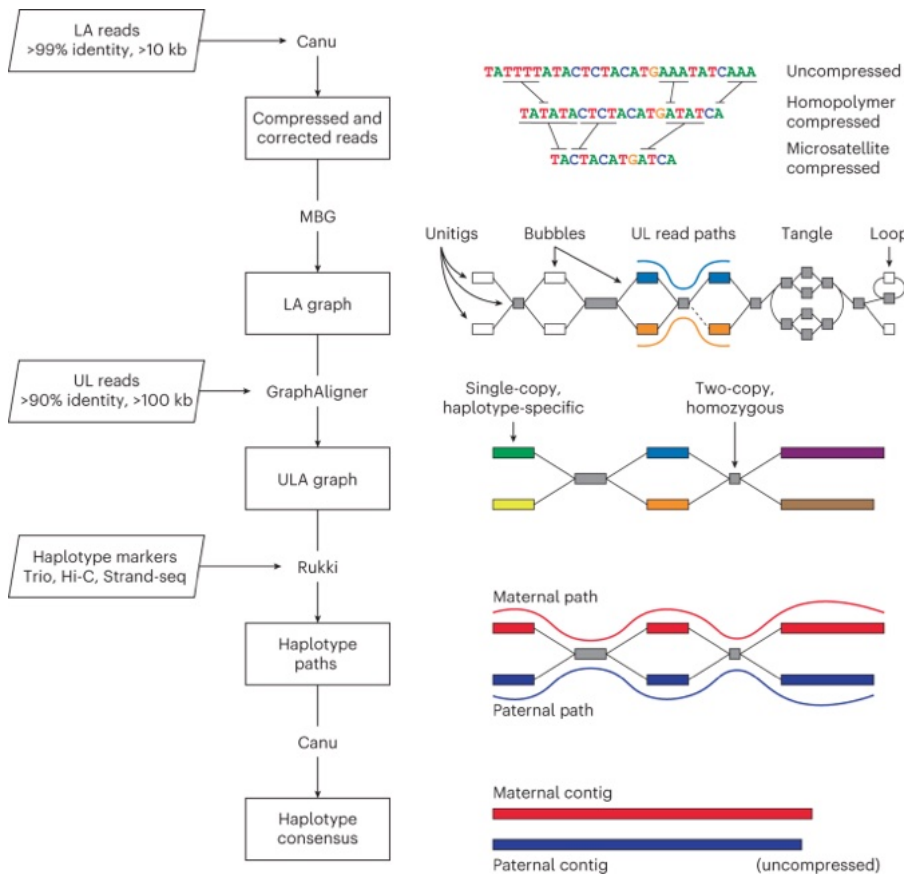
VOL. 376, NO. 6588 • 31 APR 2022

SPECIAL ISSUE RESEARCH ARTICLE

The complete sequence of a human genome

BY SERGEY MURK, SERGEY KOREN, ARAND RHE, MIKKO RAUTIAINEN, ANDREY V. BZKAJDE, ALLA MIKHAILOV, MITCHELL R. VOLLGER, NICOLAS ALTEMOSE, LEV URALSKY, ARIEL GERSHMAN, [...] ADAM M. PHILLIPPY

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Telomere-to-telomere assembly of diploid chromosomes with Verkko

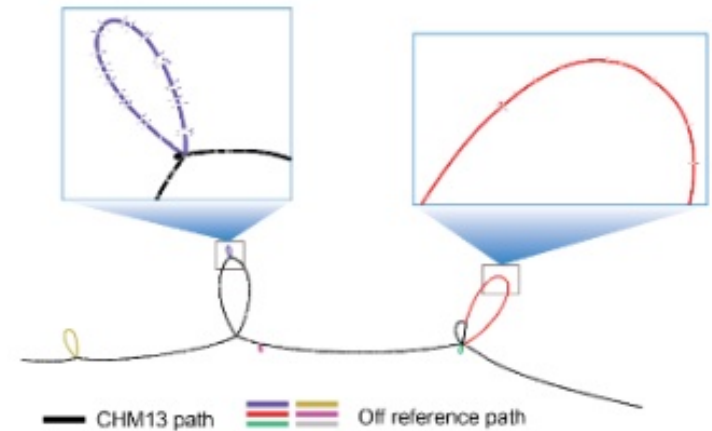
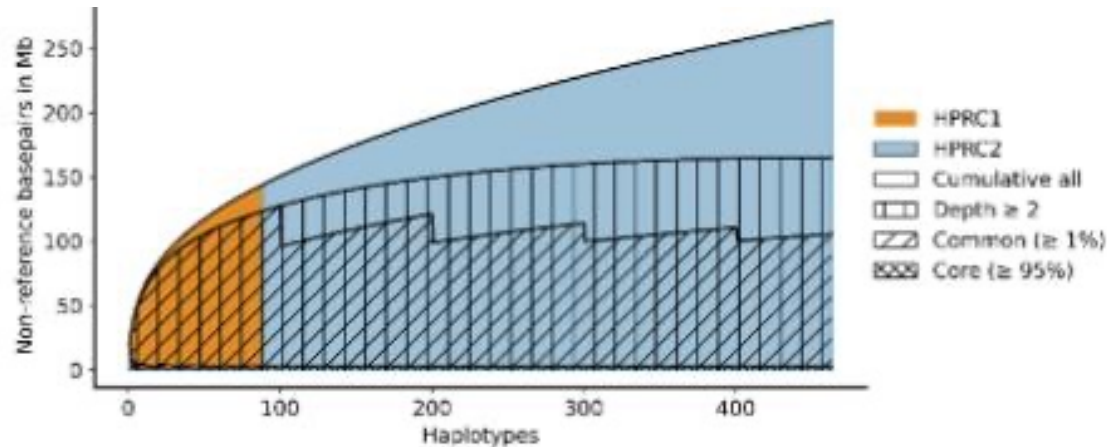
Rautiainen et al. (2023) *Nature Biotechnology* <https://doi.org/10.1038/s41587-023-01662-6>



Human Pangenome Reference Consortium (HPRC)

HPRC2 contributes 460 haplotypes that together capture over 99% of common variation

HPRC2 pangenome captures 2.6 million SNPs, 2.3 million indels and 80 thousand SVs that are off reference



HPRC2: A human pangenome reference with near-complete coverage of common genetic variation

Lucas *et al.* (2026) *bioRxiv*. <https://doi.org/10.64898/2026.07.21.739710>